General Physics Laboratory

Report 1

DETERMINING DENSITY OF RIGID OBJECTS

Class: CC10 / Group:1		
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I. Aims/Purposes:

- Determining the volume of a solid sample with a symmetrical form by measuring its size.
- Recognize the experiment's theoretical foundation, the methods used to conduct it, and the construction and functioning of the experimental apparatus.
- Recognize how to perform an experiment to determine the density of rigid objects.
- Preparing the report and calculating errors as requested.

II. Method, Equipment and Procedure:

a) Method:

- We will measure all dimensions of the given object to calculate its volume based on its symmetrical figures.

b) Equipment:

- 1 vernier caliper 0 – 150 mm with precision 0.02 mm.
- 2 measuring samples (copper ring, steel ball).

c) Procedure:

- Measuring size of the copper ring:
  + Measure the outer diameter, the inner diameter, and the height of the ring by using a vernier caliper.
  + Perform the measurement 3 times at different positions.
- Measuring size of the steel ball:
  + Measure the diameter of the steel ball by using a vernier caliper.
  + Perform the measurement 3 times at different positions.

- Measuring the mass:
  + Measure mass by using a balance consisting of a beam and two scale pans.
  + Place the object on the left pan and place the weights on right pan. Raise the beam from left by using alternative knob. Adjust the screw to make the beam balanced. (Remember to balance the beam before placing the object and weights, lower the beam when putting in or out any object.)
  + Perform the measurement 3 times for each object.

III. Equations:

a) Measuring the volume of the objects

- Determining the volume of the copper ring:

$$V = \frac{\pi}{4}.(D^2 - d^2).h$$

D: the outer radius the copper ring
d: the inner radius the copper ring
h: the height of the copper ring

- Determining the volume of the steel ball:

$$V = \frac{1}{6}.\pi.d^3$$

d: the radius of the ball

b) Measuring the density of the objects

- Determining the density:

$$\rho = \frac{m}{V}$$

m: the mass of the object
V: the volume of the object

- Determining the error:

$$\frac{\Delta\rho}{\overline{\rho}} = \frac{\Delta m}{\overline{m}} + \frac{\Delta V}{\overline{V}}$$

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IV. Experimental Data:

Table 1 (for copper ring)								
Experiment	D (mm)	ΔD (mm)	d (mm)	Δd (mm)	h (mm)	Δh (mm)	m (g)	Δm (g)
1	34.92	0.33	25.08	0.01	6.78	0.03	25.96	0.01
2	34.94	0.35	25.06	0.03	6.74	0.01	25.98	0.01
3	33.90	0.69	25.12	0.03	6.72	0.03	25.96	0.01
Average	34.59	0.46	25.09	0.02	6.75	0.02	25.97	0.01

(Accuracy of Caliper: 0.02 mm)

Table 2 (for steel ball)				
Experiment	d (mm)	Δd (mm)	m (g)	Δm (g)
1	7.98	0.01	2.08	0.01
2	7.98	0.01	2.08	0.01
3	8.00	0.01	2.06	0.01
Average	7.99	0.01	2.07	0.01

(Accuracy of Caliper: 0.02 mm. Accuracy of counter balance:0.02 g)

V. Calculations

V.1 Calculating the absolute errors of the measurement of dimensions

$$\Delta x = \Delta x_{instrument} + \Delta\overline{x}\ (m)$$

- *For copper ring:*

$\Delta D = \Delta D_{sys} + \Delta\overline{D} = 0.02 + 0.46 = 0.48 \times 10^{-3}\ (m)$

$\Delta d = \Delta d_{sys} + \Delta\overline{d} = 0.02 + 0.02 = 0.04 \times 10^{-3}\ (m)$

$\Delta h = \Delta h_{sys} + \Delta\overline{h} = 0.02 + 0.02 = 0.04 \times 10^{-3}\ (m)$

- *For steel ball:*

$\Delta d = \Delta d_{sys} + \Delta\overline{d} = 0.02 + 0.01 = 0.03 \times 10^{-3}\ (m)$

V.2 Calculating the total absolute errors of the measurement of mass m

$$\Delta m = \Delta m_{sys} + \Delta\overline{m}\ (kg)$$

- *For copper ring:* $\Delta m = \Delta m_{sys} + \Delta\overline{m} = 0.02 + 0.01 = 0.03 \times 10^{-3}\ (kg)$
- *For steel ball:* $\Delta m = \Delta m_{sys} + \Delta\overline{m} = 0.02 + 0.01 = 0.03 \times 10^{-3}\ (kg)$

V.3 Calculating the density of rigid objects

($\pi = 3.14$ and $\Delta\pi = 0.005$)

- *For copper ring:*

$\overline{V}_1 = \frac{\pi}{4}\left(\overline{D}^2 - \overline{d}^2\right)\overline{h} = \frac{\pi}{4}(34.59^2 - 25.09^2) \times 6.75 \approx 3005.70 \times 10^{-9}\ (m^3)$

And

$$\delta = \frac{\Delta V_1}{\overline{V}_1} = \frac{\Delta\pi}{\pi} + \frac{2.(\overline{D}.\Delta D + \overline{d}.\Delta d)}{\overline{D}^2 - \overline{d}^2} + \frac{\Delta h}{\overline{h}} = \frac{0.005}{3.14} + \frac{2 \times (34.59 \times 0.48 + 25.09 \times 0.04)}{34.59^2 - 25.09^2} + \frac{0.04}{6.75} \approx 0.0625$$

=> $\Delta V_1 = \delta.\overline{V}_1 = 0.0625 \times 3005.70 \times 10^{-9} \approx 187.856 \times 10^{-9} m^3$

- *For steel ball:*

$\overline{V}_2 = \frac{1}{6}\pi\overline{d}^3 = \frac{1}{6} \times 3.14 \times 7.99^3 \approx 266.94 \times 10^{-9} (m^3)$

$\delta = \frac{\Delta V_2}{\overline{V}_2} = \frac{\Delta\pi}{\pi} + \frac{3\Delta d}{\overline{d}} = \frac{0.005}{3.14} + \frac{3 \times 0.03}{7.99} \approx 0.013$

=> $\Delta V_2 = \delta.\overline{V}_2 = 0.013 \times 266.94 \times 10^{-9} \approx 3.43 \times 10^{-9} (m^3)$

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- *For calculate the density:*
  + Copper ring:

$$\overline{\rho}_1 = \frac{\overline{m}}{\overline{V}_1} = \frac{25.97\text{x}10^{-3}}{3005.70\ \text{x}10^{-9}} \approx 8640.25\ (\frac{kg}{m^3})$$

$$\gamma = \frac{\Delta\rho_1}{\overline{\rho}_1} = \frac{\Delta m}{\overline{m}} + \frac{\Delta V_1}{\overline{V}_1} = \frac{0.03\text{x}10^{-3}}{25.97\text{x}10^{-3}} + \frac{187.856\ \text{x}10^{-9}}{3005.70\ \text{x}10-9} \approx 0.064$$

=> $\Delta\rho_1 = \overline{\rho}_1 . \gamma = 8640.25 \text{ x } 0.064 \approx 552.98\ (\frac{kg}{m^3})$

  + Steel ball:

$$\overline{\rho}_2 = \frac{\overline{m}}{\overline{V}_2} = \frac{2.07\text{x}10^{-3}}{266.94\ \text{x}10^{-9}} \approx 7754.55\ (\frac{kg}{m})$$

$$\gamma = \frac{\Delta\rho_2}{\overline{\rho}_2} = \frac{\Delta m}{\overline{m}} + \frac{\Delta V_2}{\overline{V}_2} = \frac{0.03\text{x}10^{-3}}{2.07\text{x}10^{-3}} + \frac{3.43\text{x}10^{-9}}{266.94\ \text{x}10^{-9}} \approx 0.027$$

=> $\Delta\rho_2 = \overline{\rho}_2 . \gamma = 7754.55 \text{ x } 0.027 \approx 209.37\ (\frac{kg}{m^3})$

VI. Conclusions:

- *For copper ring:* $\rho_1 = (\overline{\rho}_1 \pm \Delta\rho_1) = (8640.25 \pm 552.98)\ (\frac{kg}{m^3})$
- *For steel ball:* $\rho_2 = (\overline{\rho}_2 \pm \Delta\rho_2) = (7754.55 \pm 209.37)\ (\frac{kg}{m^3})$

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